INCH-POUND
MIL-DTL-24643/44E
22 August 2002
SUPERSEDING
MIL-C-24643/44D
22 November 1994

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, 600 VOLT, TYPE LS3SF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24643 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First -	Copper conductor, AWG Size No. 18, class B stranding, ASTM B8, (7 by 0.0152 inch), uncoated.
Second -	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third -	Cross-linked polyethylene insulation. Nominal wall thickness 0.020 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements. Colored insulation, one black, one white and one red conductor for each triad.
Fourth -	Three conductors cabled together with a nominal lay of 1.5 inches to form a triad.
Fifth -	One polyester tape per NEMA FI 4.
Sixth -	Braided shield of AWG No. 34 or No. 36 uncoated or tin coated copper. Braided angle of 30 to 35 degrees. Minimum coverage of 85 percent.
Seventh -	Shield insulation of two polyester tapes per NEMA FI 4, outer tape sealed. An alternate shield insulation may consist of one polyester tape, per NEMA FI 4, plus an extruded jacket of a suitable transparent material, minimum average thickness of the transparent material 0.003 inch. The standard identification code shall be applied by method 2 on the inner tape.

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Construction (non-watertight) (continued)

Eighth - The specified number of triads (see table I) cabled together with a lay not greater than 16 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with No. 1, from the center outward. A single triad shall not be used as the central core of the cable. Fillers shall be used as necessary to form a firm, well rounded assembly.

Ninth - Binder tape, applied helically with overlap.

Tenth - Cross-linked polyolefin jacket. Minimum average wall thickness 0.080.

TABLE I. Details.

Military part no. M24643/44	Type and size	Number of triads	Overall diameter (inches)	
			min	max
-01UN	LS3SF-7	7	1.000	1.040

EXAMINATION AND TESTS:

Basic electrical:	Requirements
Conductor resistance - ohms/1000 feet at 25°C, maximum	7.06
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor, within triads	2000
Shield to shield, between triads	500
Insulation resistance - Megohms -1000 feet minimum	
Conductor to conductor	500
Shield to shield	100
Conductor to shield continuity	No failure
Group A:	
Visual and dimensional	No failure
Capacitance	
Mutual capacitance - 1 kilohertz (kHz), pF/ft, maximum	30
Capacitance unbalance - percent minimum	10
Group B:	
Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (When tested at 200 °C)	50
Cold bending, cable - at minus 30 ± 2°C,	
mandrel diameter 12 times cable diameter	No damage
Bending endurance - at 25± 2°C, cycles, minimum	25
Tear - pounds per inch thickness, minimum (ASTM D 470)	35

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Physicals (unaged)
- Insulation
  - Tensile strength - lb/in^2, minimum 1800
  - Elongation - percent, minimum 250
- Jacket (cable)
  - Tensile strength - lb/in^2, minimum 1300
  - Elongation - percent, minimum 160

Group C:

Physicals (aged)
- Insulation
  - Air oven
    - Tensile strength - percent of unaged, minimum 80
    - Elongation - percent of unaged minimum 80
- Jacket (cable)
  - Air oven
    - Tensile strength - percent of unaged, minimum 60
    - Elongation - percent of unaged minimum 60
  - Hot oil immersion
    - Tensile strength - percent of unaged, minimum 50
    - Elongation - percent of unaged, minimum 50
- Shrinkage ..... No failure
- Heat distortion - percent of unaged, maximum 30
- Permanence of printing (jacket) - cycles, minimum 125

Group D:
- Flame propagation (cable) ..... No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements
Aging and compatibility (cable) (125 ± 5°C.)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Shield insulation	2
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2

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Immersion (jacket)
- Tensile strength - percent of unaged, minimum 50
- Elongation - percent of unaged, minimum 50

Smoke index, maximum
- Jacket 25
- Fillers 45
- Insulation 45

Toxicity index, minimum
- Jacket 5
- Fillers 5
- Insulation 1.5

Durometer (jacket) - (type A) hardness, minimum....80
Weathering (jacket).... No failure

UNIT ORDERING LENGTHS: 1000 feet (nominal)

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2308-042)

Review Activities:
Army - AV, CR
Navy - CG, EC
DLA – CC

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